

Quantitative Trading Architecture

Cross-Asset Multi-Frequency Fusion Model

GRU-Based Reversal Zone Detection for Hang Seng Index

Model Structure, Mathematical Specification & Training Pipeline

1. Executive Overview

This architecture implements a two-stage deep learning system for the detection of local price extrema (reversal zones) on the Hang Seng Index (^HSI). The system first trains independent Gated Recurrent Unit (GRU) classifiers on each of five synchronized asset streams, then freezes those base models and trains a lightweight linear fusion layer that concatenates their final hidden representations. The fusion layer is supervised exclusively by the target asset's regime labels, thereby allowing sector-level peer information to refine the primary HSI decision surface.

Primary design goals are: (i) strict same-day temporal alignment across all streams, (ii) dual-frequency (micro/macro) feature representation, (iii) class-imbalance-aware training, and (iv) modular scalability to an arbitrary number of peer assets.

2. Architecture & Inference Data Flow

Figure 1 presents the complete forward computation path from raw multi-asset OHLCV streams to calibrated regime probabilities. The diagram follows the conventional machine-learning paper layout: input at the top, parallel feature extractors in the middle, fusion and output at the bottom.

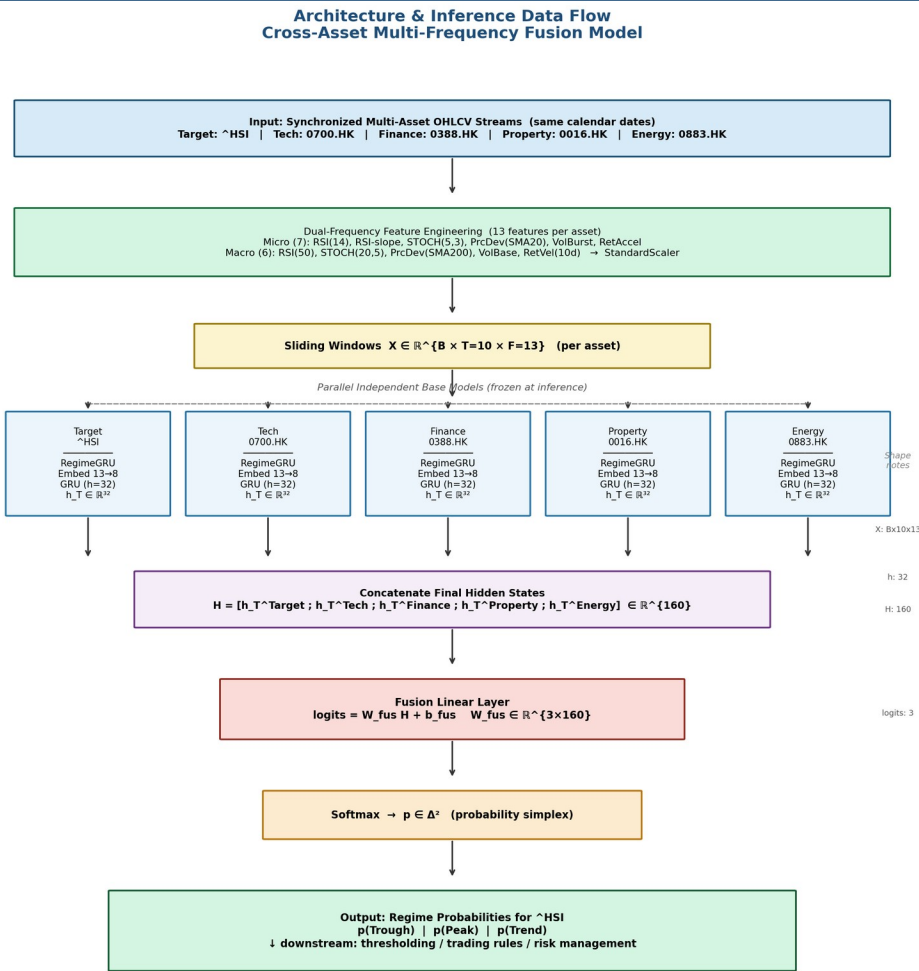


Figure 1. Architecture and inference data flow. Synchronized multi-asset streams are transformed into dual-frequency features, windowed, and processed by five independent RegimeGRU models. Their final hidden states are concatenated and passed through a linear fusion layer to produce regime probabilities for the Hang Seng Index.

3. Asset Universe and Temporal Alignment

Five equity streams are ingested over the period 2011-01-01 to 2021-01-01 (with an additional 400-day look-back for indicator warm-up). The universe is defined as:

Role	Ticker	Description
Target	^HSI	Hang Seng Index (primary trading instrument)
TechAnchor	0700.HK	Tencent – technology sector proxy
FinanceAnchor	0388.HK	HKEX – financials / exchange proxy
PropertyAnchor	0016.HK	Sun Hung Kai Properties – real-estate proxy
EnergyAnchor	0883.HK	CNOOC – energy / commodities proxy

After independent feature computation, the intersection of all five trading calendars is taken, guaranteeing that every training sample corresponds to an identical calendar date across the entire cross-section.

4. Dual-Frequency Feature Engineering

Thirteen technical indicators are extracted for every asset and partitioned into micro (short-horizon) and macro (longer-horizon) groups. All RSI and Stochastic values are scaled to the unit interval [0, 1]. Missing values after indicator calculation are filled with zero.

4.1 Micro Features (7)

- **Micro_RSI** = $\text{RSI}(\text{Close}, 14) / 100$
- **Micro_RSI_Slope** = $\Delta_3(\text{Micro_RSI})$
- **Micro_FastK / Micro_FastD** = $\text{STOCHF}(\text{High}, \text{Low}, \text{Close}; k=5, d=3) / 100$
- **Micro_Prc_Dev** = $\text{Close} / \text{SMA}(\text{Close}, 20) - 1$
- **Micro_Vol_Burst** = $\text{Volume} / \text{SMA}(\text{Volume}, 20)$
- **Micro_Ret_Accel** = $r_t - \text{SMA}(r, 5)$, where $r_t = \text{Close.pct_change}()$

4.2 Macro Features (6)

- **Macro_RSI** = $\text{RSI}(\text{Close}, 50) / 100$
- **Macro_FastK / Macro_FastD** = $\text{STOCHF}(\text{High}, \text{Low}, \text{Close}; k=20, d=5) / 100$
- **Macro_Prc_Dev** = $\text{Close} / \text{SMA}(\text{Close}, 200) - 1$
- **Macro_Vol_Base** = $\text{SMA}(\text{Volume}, 20) / \text{SMA}(\text{Volume}, 100)$
- **Macro_Ret_Vel** = $\text{Close}_t / \text{Close}_{\{t-10\}} - 1$

The resulting feature matrix for each asset is windowed into sequences of length $T = 10$. A StandardScaler is fitted exclusively on the training portion of each asset and then applied to its validation and test windows.

5. Ground-Truth Regime Labels

Labels are generated independently for every asset by a two-stage extrema detection procedure:

1. **Coarse detection:** A centred SMA of length 7 is computed. Local peaks and troughs are located with `scipy.signal.find_peaks` using a minimum distance of 7 bars and a prominence threshold equal to 3 % of the mean centred SMA value.
2. **Fine refinement:** Within a ± 1 -bar neighbourhood of each coarse extremum the exact price maximum (peak, label = 1) or minimum (trough, label = 0) is selected.
3. **Dynamic padding:** A symmetric pad of at most one bar is applied around each refined extremum, the pad length being constrained by the distance to neighbouring extrema. All remaining bars receive the default label 2 ("Trend").

Consequently the supervised task is a three-class classification problem: Trough (0), Peak (1), Trend (2).

6. Base Model – RegimeGRU

An identical architecture is instantiated for each of the five assets and trained independently on that asset's own feature sequences and regime labels.

6.1 Architecture

Let $x \in \mathbb{R}^{B \times T \times F}$ with $T = 10$ and $F = 13$.

1. Linear bottleneck embedding

$$\begin{aligned} \text{bottleneck_dim} &= \max(4, \lfloor F/2 \rfloor + 2) = 8 \\ e_t &= \text{ReLU}(W_e x_t + b_e), \quad W_e \in \mathbb{R}^{8 \times 13} \end{aligned}$$

2. Single-layer GRU

$h_t = \text{GRU}(e_1 \dots e_T; \text{hidden_dim} = 32)$

Only the final hidden state $h_T \in \mathbb{R}^{32}$ is retained.

3. Classification head

$\text{logits} = W_f h_T + b_f, \quad W_f \in \mathbb{R}^{3 \times 32}$

The forward method returns both the logits and the final hidden state; the latter is required by the subsequent fusion stage.

6.2 Training Protocol (Stage 1)

- Loss: weighted cross-entropy, weights = $N / (C \cdot n_c)$ (inverse class frequency).
- Optimiser: Adam, learning rate 1×10^{-3} , weight decay 1×10^{-4} .
- Batch size 96, 150 epochs, full-batch gradient updates on GPU when available.
- After training the model is set to evaluation mode and all parameters are frozen for the fusion stage.

7. Cross-Asset Fusion Model

The fusion stage aggregates information from all five frozen RegimeGRU models to produce a refined regime prediction for the Target (^HSI).

7.1 Architecture

Let $A = \{\text{Target}, \text{TechAnchor}, \text{FinanceAnchor}, \text{PropertyAnchor}, \text{EnergyAnchor}\}$.

For each asset $a \in A$ the corresponding base model extracts its final hidden state:

$h_T^{(a)} \in \mathbb{R}^{32}$

These vectors are concatenated:

$H = \oplus_{a \in A} h_T^{(a)} \in \mathbb{R}^{160}$

A single linear layer maps the concatenated representation to class logits:

$\text{logits_fusion} = W_{\text{fus}} H + b_{\text{fus}}, \quad W_{\text{fus}} \in \mathbb{R}^{3 \times 160}$

Class probabilities are obtained by the standard softmax transform. Only the parameters of the fusion layer ($W_{\text{fus}}, b_{\text{fus}}$) are trainable; all base-model parameters remain frozen.

7.2 Training Protocol (Stage 2)

- Supervision is supplied exclusively by the Target asset's regime labels.
- Identical weighted cross-entropy loss and Adam optimiser ($\text{lr} = 1 \times 10^{-3}$, weight decay = 1×10^{-4}).
- 100 epochs, batch size 96.
- Because the base GRUs are frozen, the computational cost of fusion training is modest.

8. Complete Training Pipeline

Figure 2 summarises the two-stage optimisation procedure. Stage 1 trains each RegimeGRU independently on its own asset's labels. After convergence the base parameters are frozen and Stage 2 trains only the fusion linear layer under Target supervision. This design is common in multi-view / multi-modal learning literature: specialised extractors are first learned, then a lightweight aggregator is optimised.

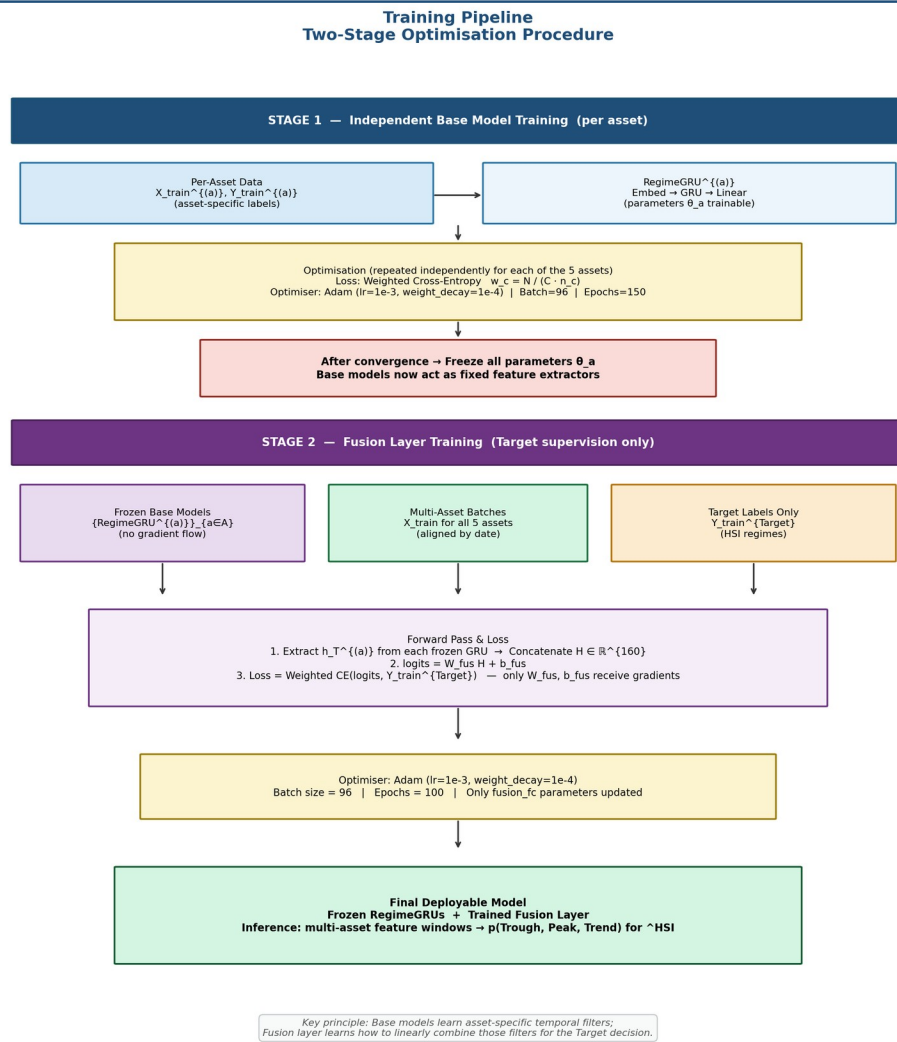


Figure 2. Training pipeline. Stage 1 performs independent supervised training of five RegimeGRU models. After freezing, Stage 2 trains only the fusion layer using concatenated hidden states and the Target asset's regime labels.

9. Temporal Data Partition

After windowing, the chronological sequence of samples is partitioned as follows:

Partition	Length (days)	Purpose
Training	Remainder after val+test	Parameter estimation
Validation	504 ($\approx$ 2 years)	Hyper-parameter / threshold search
Test	504 ($\approx$ 2 years)	Final out-of-sample evaluation

No look-ahead leakage is possible because scaling statistics are computed solely on the training windows and the chronological order is strictly preserved.

10. Inference and Downstream Usage

At inference time a dictionary of feature tensors (one per asset) is passed to the fusion model. Softmax probabilities for the three regimes are returned. These probabilities serve as inputs to a subsequent validation grid-search that optimises trading-rule thresholds (buy/sell confidence cut-offs, holding

constraints, etc.). Partial-dependence-style two-dimensional heat-maps are also generated to visualise the decision surface of the fusion model over pairs of micro features.

11. End-to-End Mathematical Flow

For a given calendar date t the complete forward computation is:

1. For each asset a : extract the 10×13 feature window $X^{\{a\}}_t$
2. $e^{\{a\}} = \text{ReLU}(W_e^{\{a\}} X^{\{a\}}_t + b_e^{\{a\}})$
3. $h_T^{\{a\}} = \text{GRU}^{\{a\}}(e^{\{a\}})_{\text{final}}$
4. $H = [h_T^{\text{Target}}; h_T^{\text{Tech}}; h_T^{\text{Finance}}; h_T^{\text{Property}}; h_T^{\text{Energy}}]$
5. $\text{logits} = W_{\text{fus}} H + b_{\text{fus}}$
6. $p = \text{softmax}(\text{logits}) \in \Delta^2$ (probability simplex over {Trough, Peak, Trend})

The resulting probability vector p constitutes the model's calibrated assessment of the likelihood that date t belongs to a local trough, a local peak, or a non-extremal trend regime on the Hang Seng Index.

12. Design Rationale and Limitations

Freezing the base GRUs after individual training prevents the fusion stage from overwriting asset-specific temporal filters while still permitting the linear layer to learn optimal linear combinations of the five hidden states. The dual-frequency feature set supplies both short-term momentum/volume signals and longer-term structural context. The same-day intersection constraint eliminates asynchronous calendar effects that would otherwise introduce spurious cross-asset correlations.

Limitations include the relatively short sequence length ($T = 10$), the modest capacity of a single-layer GRU, and the assumption that sector proxies remain informative over the entire sample. Future extensions may incorporate attention-based fusion, multi-horizon labels, or online adaptation of the fusion weights.

— End of Model Specification —